



Impact of artificial intelligence investment on firm value

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Abstract

As artificial intelligence (AI) has recently gained momentum and attention, the interest and investment in AI have also accelerated. However, the impact of AI on firm value is rarely discussed. On the basis of the 119 announcements of 62 listed firms who have invested in AI, this study finds that AI investment has a negative impact on the firms' market value. The stock prices of the firms decrease by 1.77% on the day of the announcement. Nonmanufacturing firms and firms with weak information technology capabilities or low credit ratings suffer a more negative impact compared with other firms. The findings suggest that investors perceive AI investment announcements to be unwelcome news for the majority of firms. Subsequently, the characteristics affecting the shareholders' reaction towards AI adoption are presented. This research offers one of the first empirical evidence about the market value of AI and provides a reference for firms interested in investing in AI.

Keywords Artificial intelligence · AI · Event study · Market value · Contextual factors

1 Introduction

Artificial intelligence (AI) is a disruptive technology. Companies across sectors have increasingly harnessed AI's power in firm operations, and managers have made major investments in adopting AI to improve their operations. For instance, in the banking industry, Bank of America has used a virtual assistant called Erica to help customers handle basic banking

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tasks while the Bank of New York Mellon Corp has implemented more than 220 “bots” to handle repetitive tasks (Castellanos 2018). Analysts suggest that AI can help the banking industry save more than US\$1 trillion by 2030 by reducing 22% of their firms’ operating expenses (Joyce 2018). McKinsey Global Institute also has reported that AI can provide an additional \$13 trillion per year to the global economic output by 2030 (Bughin et al. 2018). According to a survey conducted in 2016 by Statista (2016), a provider of market research and research and analysis services, revenues from AI for enterprise applications can grow from \$1.62 billion in 2018 to \$31.2 billion in 2025. Moreover, 84% of enterprises believe that AI implementation will spur more competitive edges, and 75% of them believe that AI implementation will lead to new business opportunities. In addition, according to a study published by Fortune (Murry 2017), more than 80% of Fortune 500 Chief Executive Officers (CEOs) believe that AI is extremely critical to their firms’ futures.

Despite all the hype, some companies remain concerned about the technologies and hesitate to use them. Firms worry that machines are not armed with functions that mirror the same goals as those of the management, and this gap may lead to incorrect decisions or cause fatal mistakes in operations. First, AI adoption will likely be a substantial challenge for organizations since they may experience reputational damage, revenue losses, and diminished public trust if the AI machine falters. Second, costs are another issue that may prevent firms from using AI technology. Apart from the costs associated with the implementation (i.e., system costs, project and consulting fees, system integration costs, and system upgrading costs), various other costs will likely be incurred when organizations transit to AI (Bughin et al. 2018). The transition costs may include those costs incurred while restructuring the organization, severance for workers who may be displaced by the new technology, and the costs to upgrade the skills of existing employees (Bughin et al. 2018). Therefore, the impact of AI on firms may not be always positive as people expect them to be.

The objective of this research is to provide timely and objective evidence of the value of AI investment. The investor perspective is considered to examine the value of AI. The event study methodology is employed to examine the effect of AI investment on market value. Since a stock market response is instantaneous and can reflect investors’ expectations of future firm performance (Bose and Pal 2012), an event study is an appropriate approach for this research. The event study is also widely cited in the literature as a means of investigating how information technology (IT) investments affect firm value. Some previous studies have shown that IT investments are positively correlated with the market value of firms (Dehning et al. 2003; Subramani and Walden 2001), whereas other studies have reported a negative relationship (Bose et al. 2011). Thus, we have searched and analyzed relevant articles from 2015 to 2019 to identify events that can reveal information about a firm’s initiative in adopting AI. The rest of the study is organized as follows. Section 2 provides the theoretical background, and Sect. 3 enumerates the hypotheses. Section 4 presents the research method, and Sect. 5 presents the result. The result of our investigation and the implications are discussed in Sect. 6.

2 Theoretical background

2.1 AI as a disruptive technology in operations management

As a general term, AI refers to computer systems that perform tasks that usually require human intelligence (Mueller and Massaron 2018). AI is specifically referred but not limited

to different types of technologies, such as expert systems, natural language processing, virtual/intelligent agents, robotic process automation, speech recognition and machine vision, artificial neural networks, etc. Moreover, AI is commonly regarded a disruptive technology since it can disrupt operations in different industries, including agriculture (Bannerjee et al. 2018), manufacturing (Benotmane et al. 2019), banking (Fethi and Pasiouras 2010), and healthcare (Fan et al. 2020; Jiang et al. 2017). Since the 1980s, numerous studies have attempted to investigate and solve problems in operations management (OM) by using AI (Kobbacy et al. 2007).

According to two detailed literature reviews conducted by Kobbacy and Vadera (2011) and Kobbacy et al. (2007), over 1200 papers from 1995 to 2004 and over 1400 papers from 2005 to 2009 on the application of AI technologies in the areas of OM were published. AI is employed in the following four areas of OM: design; scheduling; planning and control; and quality, maintenance, and fault diagnosis (Kobbacy and Vadera 2011; Kobbacy et al. 2007). The commonly used AI technologies include case-based reasoning (CBR), genetic algorithms (GAs), neural networks (NNs), knowledge-based systems, fuzzy logic, and hybrid approaches (Kobbacy and Vadera 2011). Before 2005, fuzzy logic was heavily used in design and scheduling, whereas NNs dominated process planning and control (Kobbacy et al. 2007). From 2003 to 2004, a dramatic increase in the use of GAs was found for design and scheduling. From 2001 to 2002, the use of NNs dominated and increased significantly for the other two aspects (Kobbacy and Vadera 2011).

With the rapid development of computing and networking technologies in recent years, the number of studies on AI in the OM domain has increased. Chou et al. (2010) presented and compared multiple regression analysis, artificial NNs, CBR, and hybrid AI models to predict the costs of the equipment used to manufacture thin-film transistor liquid-crystal displays. Munguia et al. (2010) proposed a system that uses several AI technologies, such as fuzzy logic, to guide the design of the optimum selection of production parameters according to the product requirements in the early design stages. Lawrynowicz (2011) used GAs for job shop scheduling in supply networks and collective scheduling in an industrial cluster. McNaught and Chan (2011) described a Bayesian network-based intelligent decision support system that helped operators in Motorola diagnose and correct faults during product system testing. With the aim of reducing emissions from waiting trucks and improving crane operations at container terminals, Do et al. (2016) developed a method that used a discrete event simulation to measure the total truck waiting time and crane moving distance and then applied a GA to reduce the emissions produced from these trucks and cranes. Kalayci et al. (2016) suggested a hybrid algorithm that combines a GA with a variable neighborhood search method to solve the sequence-dependent disassembly line balancing problem in reverse supply chains. Aouadni and Rebai (2017) proposed a decision support system based on the multi-criteria satisfaction analysis method and the continuous GA and multi-objective GA to help organizations evaluate and measure their employees' job satisfaction. Chen et al. (2020) presented a multi-objective GA for energy-efficient hybrid flow shop scheduling with lot streaming to reduce the completion time and energy costs at the manufacturing-system level. Yang et al. (2019) proposed a grey-box model based on the GA to help shipping companies predict fuel consumption and showed that the proposed model is remarkably better than the latest grey-box model. More recently, Dosdoğru et al. (2020) developed a new hybrid AI-based decision support framework to help resolve the inventory routing problem.

2.2 Impact of AI investment on a firm

With the adoption of AI technologies becoming a leading trend in most companies to manage their operations in recent decades, the interest of researchers has focused on investigating whether AI technologies can ultimately bring benefits and value to companies or impact them altogether. To the best of our knowledge, the number of studies that utilized qualitative and quantitative approaches to assess the impact of AI on firms in practice, generated through questionnaires and interviews with leaders and practitioners in specific industries, is extremely limited. Jain (2019) collected data from 50 selected business decision-makers and regular employees engaged in Indian firms through an online survey, and the results of the analysis showed a significant impact of AI on the economic growth of their businesses. In a recent study about the impact of AI on pharmaceutical businesses, the results of a questionnaire indicate that AI has a positive effect on the performance of such business in Thailand (Chetthamrongchai and Jernsittiparsert 2020). In a survey conducted by Basri (2020), AI-assisted social media marketing helped increase the number of customers and customer bases and improved firm profitability. In a recently published work, Minevich (2020) reported the process of how major players in some industries used and benefitted from their AI projects. For example, Domino's Pizza has reduced delivery time and improved delivery time prediction accuracy from 75 to 95%, and mining companies in Australia have cut mining costs, improved worker safety, and boosted productivity by 20% using autonomous trucks and drilling technology. However, no academic research has quantified the economic values created for firms through the adoption of AI. Obviously, we have identified a huge gap in the AI and OM research literature that has not yet been explored.

2.3 IT's influence on firms' market value

Previous studies have used the event study methodology to examine the impact of various types of IT investments on firm value (e.g., Dehning et al. 2003; Teo et al. 2016). Previous studies show inconclusive findings, and no insignificant, negative, and positive relationships were shown between IT investment and the market value of a firm. Studies in this field suggest that the effects of IT investment on market value are contingent upon various contextual factors, including the characteristics of the adopting firms, the IT investment type, and the industry type. Table 1 summarizes the IT investment-related event studies.

3 Hypotheses

The prediction of the costs and benefits of applying AI determines the effect of the AI investment on the market value of a firm. The market will weigh the discounted value of the long-term benefits brought by the AI investment and the transitory costs. The market value will grow if the expected long-term benefits are higher than the costs foreseen for the investment and vice versa.

Investing in an emerging technology, such as AI, will generally increase the firm's expenditures and assets but reduce its cash flow. Moreover, AI usually involves huge investments and high risk, and its benefits and payback time are difficult to measure. Other than the costs associated with the implementation, such as the system costs, project and consulting fees, and system integration and system upgrading expenses, other various costs are likely to be incurred when organizations transit to AI (Bughin et al. 2018). Transition costs may include

Table 1 Prior event studies examining stock market reaction to IT investments announcements

References	Investment	Key findings
Dos Santos et al. (1993)	IT Investment	No abnormal returns on full sample 1.03% returns on innovative IT investment announcements
Chatterjee et al. (2002)	IT Investment	1.99% returns on full sample 2.01% returns on Infrastructure announcements 0.84% returns on application announcements
Hunter (2003)	IT investment	– 0.85% returns for retail firms – 0.68% returns on exploitative IT investment – 1.11% returns on exploratory IT investment No support found for size, slack, and time
Bose et al. (2011)	RFID	– 0.29% on full sample US-based, low growth potential, and less diversified firms experience more negative impact
Teo et al. (2016)	Business analytics	Positive return on full sample More positive to firms with high sales growth and high return on assets

the costs incurred while restructuring the organization, severance if some workers may be displaced by new technologies, and the costs to upgrade the skills of existing employees (Bughin et al. 2018). In addition, in a recent survey involving 29 company leaders from AI-related companies, only 48% of them felt positive about their AI startup. Those AI-related companies cannot easily prove that AI is beneficial to their company (Faggella 2019). Therefore, investors may observe that the firms investing in disruptive technology, including AI, are financially unattractive in the short term and thus may not choose the stocks of those firms (Staw et al. 1981).

Nonetheless, information suggests that technology will bring long-term advantages, and the costs are temporary in nature. Thus, the market may consider AI adoption as positive news for a firm. In the last decade, the research and surveys conducted by different organizations have suggested that AI is important and good for companies. International companies, such as Amazon, Apple, and Facebook, are using AI as part of the business to automate their operations successfully (Marr 2019). In addition, major players in different industries reported their use of and benefits from AI. Domino's Pizza has used AI to reduce delivery times and predict delivery times more accurately while Barclay's has used AI to detect fraud and improve the customer experience through chatbots (Minevich 2020). An analyst also predicted that the cost reduction in the operating expenses of financial institutions can reach 22% after using AI (Joyce 2018). This kind of good news is notably impressive to investors and enables them to have great expectations of AI. Hence, investors may observe the firms that claim they are implementing AI attractive and may consider investing in them.

Since AI implementation will have great impacts on the costs, risks, and revenues of companies, investors will pay much attention on this technology and consider AI implementation as one of the factors prior to making investment decisions. Some investors may regard AI adoption as good news, but other investors may stay away from companies implementing AI. Previous research has also illustrated the significant effects, both positive (Chatterjee et al. 2002; Teo et al. 2016) and negative (Hunter 2003; Tanriverdi and Ruefli 2004), of innovative IT on the market value of a firm. Therefore, we hypothesize the following:

H1 The announcement of AI investment has a significant impact on a firm's market value.

IT capability is defined as an organization's ability to identify information systems to meet business needs, deploy these systems in a cost-effective manner, and provide long-term maintenance and support for the systems (Karimi et al. 2007; Ross et al. 1996). IT capabilities, including IT infrastructure capability and IT business spanning capability, enhance organizational agility, such as operational adjustment agility (Lu and Ramamurthy 2011). Prior research has found the association of IT capability with firm performance (Lu and Ramamurthy 2011; Stoel and Muhanna 2009). Firms with a strong IT capability have better IT expertise and IT infrastructure and thus are more likely to successfully implement new IT technology than firms with weak IT capability. However, firms with a weak IT capability give less effort to IT innovations compared with their counterparts with a strong IT capability. Therefore, the firms with a weak IT capability are less capable of making the best use of IT innovation or less likely to successfully implement new technology, such as AI. Thus, we hypothesize the following:

H2 Firms with a weak IT capability experience a more negative impact on their market value than firms with a strong IT capability.

Financial healthiness is an important factor in determining the success of IT adoption (Hayes et al. 2001). Firms may need to commit large amounts of money to acquire physical and human resources for the implementation of AI (Bughin et al. 2018). Hence, sufficient resources are needed for successful AI implementation. However, the expenses required in adopting AI may increase the debt or reduce the cash flow of the company. Financially unhealthy companies are expected to suffer more negative effect of AI investment than financially healthy firms. If the return of the AI investment cannot cover the costs of investing in the technology, then the financially unhealthy firm is likely to experience a poor cash flow. A third-party credit rating can be used to measure a firm's financial healthiness (White 2010). A low credit rating suggests the difficulty of a firm to pay its debts, whereas a high credit rating indicates the financial strength of a firm and its ability to pay its debts. We can consider low-credit rating firms as financially unhealthy companies and high-credit rating firms as financially healthy companies. Previous work has also studied investors' responses to credit rating announcements (Castellano and D'Ecclesia 2013). Therefore, the market may observe low-credit rating (financially unhealthy) firms as more likely to experience the negative impacts of AI adoption than high-credit rating (financially healthy) companies. Hence, we suggest the following hypothesis:

H3 Firms with low credit ratings experience a more negative impact on their market value than firms with high credit ratings.

The link between AI investment and industry type is unclear and has not yet been extensively studied. Previous studies suggest that industry type affects the influence of IT investment on firm performance (Dos Santos et al. 1993; Kohli and Devaraj 2003; Sohal et al. 2001). Ranganathan and Brown (2006) determined that ERP platforms provide more benefits to firms in manufacturing industries. In the context of AI adoption, investors may perceive AI adoption as more potentially beneficial for firms in the manufacturing industry because operational performance, such as labor productivity and quality improvement, seems to be more potentially enhanced by AI than other companies (Bughin et al. 2018; Chui et al. 2018). In particular, investors may perceive AI adoption to be less beneficial to firms in nonmanufacturing industries (Bughin et al. 2018). Nonmanufacturers with AI adoption may not be able to cover the costs of the AI adoption, and enhancements in productivity and profitability are difficult to observe. Therefore, we make the following hypothesis:

H4 Firms in nonmanufacturing industries experience a more negative impact on their market value than firms in manufacturing industries.

4 Research methodology

We initially adopted the event study methodology, which is popular in the field of finance and accounting, to estimate the stock market reaction to AI investment announcements, a phenomenon that has become increasingly common in the context of OM and IT investments (Bose and Leung 2014; Bose and Pal 2012; Lam et al. 2019). The event study produces reliable results even under less perfect conditions, and it allows investors to quantify the instantaneous effect of an activity implemented by a firm instead of waiting for the effect estimation of its accounting measures, which may only be made available after several months (Bose and Pal 2012). First, we collected news announcements related to the AI investments of firms. The collected data were filtered and processed before the analysis. Subsequently, we estimated the expected return by using a regression model and calculated the abnormal returns (ARs) and cumulative abnormal returns (CARs) in the event window. The significance level of the CARs was evaluated using different parametric and nonparametric tests. Finally, we split the data into subsamples for further analysis. The ARs of the subsamples were computed, and the significance of the various factors in the subsampling was analyzed.

4.1 Data collection and processing

We used Factiva to search for announcements containing pertinent keywords, such as “AI” and “Artificial Intelligence,” from January 2015 to February 2019 and subsequently collected the news announcements related to AI investments by the firms. Each announcement was carefully examined to determine its relatedness to the AI investments of firms. We eliminated announcements that were irrelevant or related to nonlisted firms. Only the earliest announcement was selected when more than one announcement of the same event was found. We also removed announcements related to stocks if they have insufficient historical data or thinly traded stocks. The announcement of firms jointly engaged in AI initiatives was treated as a separate event for each firm. The announcements related to confounding news, including earnings announcements/dividends, acquisitions/mergers, top management changes, and rating changes (e.g., credit rating), within a five-day window were excluded. After a careful processing of the data, a total of 119 valid announcements representing 62 sample firms from 2015 to 2019 were maintained. Some of the announcements included in our sample are shown in Table 2.

We collected stock price information from the Center for Research in Security Prices (CSRP) and financial data from COMPUSTAT. Following previous research (Bharadwaj 2000; Lim et al. 2013; Muhanna and Stoel 2010), we determined if a firm with a high IT capability was included in the InformationWeek 500 in the announcement year or in any other year within our five-year study period. We adopted domestic long-term credit rating which is produced by Standard and Poor to indicate a firm’s credit rating (Bose and Leung 2013). The ratings equal to or above A- were classified as high credit ratings. We used SIC codes to define whether the firms were part of the manufacturing sector. A firm was considered to be a manufacturing firm if its SIC code is from 20 to 39; otherwise, it was regarded as a nonmanufacturing firm.

Table 2 Sample of announcements

Firm	Date	News
Citigroup	Feb 1 2019	Citigroup is deploying AI to monitor corporate customers' payments
Kraft Heinz	Oct 3 2017	Kraft Heinz is working with AI across all areas to identify inefficiencies and optimize performance
Mastercard	Nov 30 2016	"Mastercard today introduced Decision Intelligence which uses AI technology to help financial institutions increase the accuracy of real-time approvals of genuine transactions and reduce false declines." From Dow Jones Institutional News
Pfizer	May 3 2017	"Pfizer Australia is rolling out AI-based digital analyst tools from Complexica in a bid to improve the pharmaceutical company's data-driven sales and marketing decision making." From Complexica

Table 3 presents the description of the sample firms in our dataset. Panels A and B present how the sample firms are distributed across years and industries, respectively. Panel C shows the sample firms' financial statistics in terms of the number of employees, total assets, return on assets, cash, and long-term debt.

4.2 Abnormal stock return

We employed the capital asset pricing model (CAPM), which is the most common computation market model, to estimate abnormal stock returns. CAPM is represented by

$$E(R_{it}) = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}, \quad (1)$$

where R_{it} is the rate of return for announcement i on day t , R_{mt} is the rate of return of market index m on day t , α is the y-intercept and β_i is the slope for announcement i , and ε_{it} is the error term. We considered multiple stock indices for each stock and selected the market index with the highest R^2 for the regression model. To obtain the main market reaction corresponding to the events, the AR was estimated one day before the announcement, on the announcement day, and one day after the announcement. We used stock price data for the 200-day estimation period occurring one month before the AI investment announcement to build the regression model (Eq. 1) and calculate the expected return for each sample firm.

To calculate the abnormal return (AR) in the event window, i.e., AR_{it} , as the difference between the actual return, i.e., R_{it} , and the expected return, the formula used is shown in Eq. (2).

$$AR_{it} = R_{it} - (\sigma_i + b_i R_{mt}) \quad (2)$$

The cumulative abnormal return (CAR), which is an aggregation of the ARs for the event window, is computed using Eq. (3):

$$CAR_i = \sum_{t=-1}^{+1} AR_{it} \quad (3)$$

To assess the significance level of CAR, as illustrated by Dos Santos et al. (1993), we used the parametric Z-test and two nonparametric tests, namely, the sign test and the Corrado's rank test, to improve the robustness of the results.

Table 3 The description of the sample firms*Panel A: Distribution of the AI investment announcement by year*

Years	Number	Percentage
2015	7	5.88
2016	20	16.81
2017	37	31.09
2018	36	30.25
2019	19	15.97
Total	119	100

Panel B: Distribution of AI investment announcements by industry

SIC	Industry	Frequency	Percentage
16	Heavy Construction, Except Building Construction, Contractor	1	0.84
20	Food & Kindred Products	4	3.36
28	Chemicals & Allied Products	11	9.24
29	Petroleum Refining & Related Industries	1	0.84
30	Rubber & Miscellaneous Plastic Products	1	0.84
35	Industrial & Commercial Machinery & Computer Equipment	2	1.68
36	Electronic & Other Electrical Equipment & Components	5	4.20
37	Transportation Equipment	4	3.36
42	Motor Freight Transportation	1	0.84
48	Communications	2	1.68
52	Building Materials, Hardware, Garden Supplies & Mobile Homes	1	0.84
53	General Merchandise Stores	1	0.84
58	Eating & Drinking Places	2	1.68
59	Miscellaneous Retail	7	5.88
60	Depository Institutions	14	11.76
61	Nondepository Credit Institutions	13	10.92
62	Security & Commodity Brokers, Dealers, Exchanges & Services	10	8.40
63	Insurance Carriers	4	3.36
67	Holding & Other Investment Offices	2	1.68
73	Business Services	29	24.37
80	Health Services	1	0.84
87	Nonclassifiable Establishments	1	0.84
99	Engineering, Accounting, Research, & Management Services	2	1.68
	Total	119	100

Panel C: Financial details of the firms making AI investment announcements

Variable	Employee no. ^a	Total assets ^b	ROA	Long term debt ^b	Cash ^b
Mean	134	396,139	0.14	44,906	11,429
Median	72	150,208	0.12	18,482	8079
Std. Deviation	681,973	232	0.12	16,012	64,710
Minimum	2	1694	− 0.25	0	165
Maximum	2200	2622,532	0.39	249,637	51,840

^aThousands^bUSD in Millions

Table 4 Overall impact of announcement of AI investment on stock price in varies event windows

	[− 1]	[0]	[1]	[− 1, 0]	[0, 1]	[− 1, 1]
Mean CAR	− 1.66%	− 1.77%	− 1.51%	− 3.44%	− 3.28%	− 4.94%
Z test	− 0.01	− 1.51	0.65	− 1.08	− 0.60	− 0.50
Z test (<i>p</i> Value)	0.49	0.07	0.26	0.14	0.27	0.31
Median CAR	0.11%	− 0.10%	− 0.04%	0.01%	− 0.12%	0.13%
No. of +ve:−ve returns	63:56	50:69	57:62	60:69	57:62	63:56
Sign test	0.70	− 1.68	− 0.40	0.15	− 0.40	0.70
Signed test (<i>p</i> value)	0.24	0.05	0.34	0.44	0.34	0.24
Corrado test	0.90	− 1.31	0.49	− 0.29	− 0.58	0.04
Corrado's Rank test (<i>p</i> value)	0.18	0.09	0.31	0.39	0.28	0.48

5 Results

Table 4 shows the CARs of all firms in the dataset that have released announcements of their AI activities. We estimate the CARs of different event periods, including specific event windows ([− 1], [0], and [1]) and multiple day event windows ([− 1, 0], [0, 1], and [− 1, 1]). Here, [− 1], [0], and [1] represent the day before, the day of, and the day after the announcement, respectively. The results show that the most significant impact appears on day [0] that the announcement is made.

Using both parametric and nonparametric tests, on day [0], the mean (− 1.77%) and median (− 0.10%) ARs are statistically significant and negative. This finding is reasonable since investors observe AI investment announcements negatively and reacts too soon after the release of the announcements. The result was not significant for the day preceding the announcement [− 1], suggesting that no information leakage occurs before the event day. Therefore, H1 is supported.

Table 5 demonstrates the findings of the subsample analysis on the event day [0], based on which H2–H4 are tested. The firm subsamples with weak IT capabilities in Panel A undergo a negative change, which is statistically significant (10%) according to both the parametric and nonparametric tests. However, firms with strong IT capabilities yield insignificant results in both tests. The result suggests that firms with weak IT capabilities produce a strong negative impact when they adopt AI. Therefore, H2 is supported. The firms with low credit ratings in Panel B have statistically significant negative mean and median CARs. Both parametric and nonparametric tests are significant at the 10% level. However, the results are insignificant for firms with high credit ratings. H3, which predicts that firms with low credit ratings produce a strong negative impact, is therefore supported by the result. Finally, a significant negative impact upon an AI investment announcement is observed among nonmanufacturing firms on the event day [0] in Panel C. The AR results are significant (10%) and negative according to both parametric and nonparametric tests for these nonmanufacturing firms. By contrast, the results are insignificant for the manufacturing firms. This finding suggests that AI investments strongly and negatively influence nonmanufacturing firms. Therefore, H4 is supported.

Table 5 Impact of AI investment on stock price across various factors on the event day [0]

			Mean	t test (<i>p</i> value)	Median	Mann–Whitney test (<i>p</i> value)
<i>Panel A: Impact of IT capability</i>						
	Low	High	Difference in subgroup (Low–High)			
Sample size	71	48				
Mean CAR	– 2.98%	0.01%	– 2.99	0.09	– 0.18	0.07
Z test (<i>p</i> value)	0.02	0.44				
Median CAR	– 0.20%	– 0.02%				
Sign test (<i>p</i> value)	0.03	0.45				
Corrado test (<i>p</i> value)	0.10	0.44				
<i>Panel B: Impact of credit rating</i>						
	Low	High	Difference in subgroup (Low–High)			
Sample size	36	83				
Mean CAR	– 0.25%	– 0.44%	0.19%	0.08	– 0.07	0.09
Z test (<i>p</i> value)	0.09	0.12				
Median CAR	– 0.15%	– 0.08%				
Sign test (<i>p</i> value)	0.06	0.20				
Corrado test (<i>p</i> value)	0.04	0.34				
<i>Panel C: Impact of type of industry</i>						
	Non-manufactory	Manufactory	Difference in subgroups (Non-manufactory–manufactory)			
Sample size	90	29				
Mean CAR	– 2.38%	0.11%				
Z test (<i>p</i> value)	0.02	0.27	– 2.49%	0.10	– 0.24	0.05
Median CAR	– 0.14%	0.10%				
Sign test (<i>p</i> value)	0.03	0.39				
Corrado test (<i>p</i> value)	0.06	0.30				

6 Discussion

The overall purpose of this research is to evaluate the impact of AI adoption on the market value of a firm. We suggest a hypothesis in which the announcement of AI adoption will result in significant abnormal market returns. In addition, we hypothesize that the AI adoption announcement will have a significant negative impact on companies with a weak IT capability, with a low credit rating, or that are part of nonmanufacturing industries. By using the event study methodology, we test our hypotheses by analyzing the stock market reactions to an AI adoption announcement from 2015 to 2019. The full sample analytical results prove the significant ARs associated with AI adoption announcements. Furthermore, our study finds

that negative ARs are strongly associated with AI adoption for firms with a weak IT capability and weak financial healthiness or in nonmanufacturing industries.

The full sample analysis also provides empirical evidence that AI investment announcements lead to negative abnormal stock returns. The result shows that AI investment has a negative impact (-1.77%) on the day of the announcement. The findings of this research align with those of previous IT-related event studies that find negative effects on the announcement day (Bose et al. 2011; Tanriverdi and Ruefli 2004). However, the results contradict most of the previous IT investment-related event studies that find positive returns (ARs from 0.18 to 8.2%) (Oh et al. 2006; Subramani and Walden 2001; Teo et al. 2016) or insignificant returns (ARs from 0.02% to 0.09%) (Dos Santos et al. 1993; Im et al. 2001).

We also investigate how contextual factors influence market response towards AI investments. We found that IT capability, credit rating, and type of industry are strong moderating factors. Firms with low IT capabilities and low credit ratings create more negative abnormal market value (-2.98% and -0.25% , respectively) than their counterparts on the announcement day. The results are reasonable since investors may not be confident in their low IT capability or low credit rating, which may affect the successful implementation of the AI technology. In addition, firms in nonmanufacturing industries have a more negative abnormal market value (-2.38%) than firms in manufacturing industries on the event day [0]. This result agrees with the view of previous reports that expect the manufacturing industry to benefit the most from AI technologies (Bughin et al. 2018; Chui et al. 2018). On the basis of the results, this study offers a number of theoretical and managerial contributions and implications, as discussed in the next subsections.

6.1 Contributions and implications

From a theoretical perspective, this study offers a number of contributions to theory and thus enriches the AI, IT, and OM literature. First, this study provides empirical evidence that AI announcement results in negative ARs. AI is believed to be extremely important to firms' futures in different industries (Murry 2017); however, does AI indeed always have positive impacts on firms? A number of papers and reports discuss the advantages and importance of AI in operations, such as the works published by Makridakis (2017), Joyce (2018) and Fragapane et al. (2020). Nonetheless, the empirical papers that support the financial benefits are lacking. Our results demonstrate the relationship between AI investment and the firm's market value. To the best of our knowledge, this study is the first one to objectively measure how AI investment influences the market value of firm adopters. In particular, this research contributes to an increasingly important part of the literature covering the impacts of AI investment on business value and the business model. Another theoretical contribution is the examination of the effects of certain contextual factors, namely, the IT capacity, the credit rating, and the type of industry on the returns of AI investment announcements. The results illustrate the importance of considering these factors in AI investment, thus providing a much broader picture of the OM and IT research areas. Third, this study allows researchers to comprehensively understand the impacts of adopting disruptive innovation on market value, thus providing theoretical guidance for future studies. Researchers can also conduct further studies to gain more insights into the influence of other disruptive innovations and other contextual factors on firm values.

This research holds important practical and managerial implications to practitioners and managers. First, our study concludes that AI investment can reduce a firm's market value. AI adopters need to pay particular attention to the release of new investments since they are

vital for investors and the market value of firms. Managers of AI adoption need to ensure that their senior management is aware of the potential negative market response towards an AI investment announcement. The awareness may help obtain a strong commitment and support from the senior management, even when short-term market losses occur. Second, on the basis of the examined contextual factors of AI investment, the subsample analysis provides a much deeper understanding of how an AI investment is likely to create market value for the adopting firm. We find that the market reacts more favorably towards manufacturing firms, but it responds more negatively to nonmanufacturing firms. Moreover, AI investments will lead to significant negative impacts on firms with a weak IT capability and low credit rating. To avoid suffering, nonmanufacturing firms and firms with a weak IT capability or low credit rating should apply appropriate risk management strategies for AI adoption. These firms may consider investing in AI only after they improve their IT capability or financial healthiness. Third, the findings of our study are particular useful and timely for operations managers in the case of AI initiatives that have recently blossomed. Without a doubt, managers will make investment decisions that maximize the value of firms. However, although managers make operational decisions by relying on accounting measures, including profitability and productivity, these statistics are normally unavailable immediately for the firm when AI initiative announcements are released. In fact, the data may not be available for many months. Additionally, while it may take years of observations to estimate the direct effect of an event on firm value, such measurements may not always reflect the actual financial effect of the event because accounting statistics can be manipulated by managers. Although the results of event analysis cannot fully measure the actual performance of AI implementation, they can reflect the concerns of investors about the adoption of AI and the risks associated with the adoption. Therefore, by using the event study methodology to measure the financial effect of an AI initiative, this research provides managers with a timely understanding of how the market reacts to AI initiatives, further supporting their needs to make decisions on these disruptive innovation investments.

6.2 Limitation and future research

Similar to previous research that used the event study methodology, the current research also has several limitations. First, the sample is limited to only U.S. publicly listed firms. Therefore, the results may not be generalizable to small and medium firms. Future studies may consider announcements from other countries, such as China, and compare how the market reaction differs across countries. Moreover, firms may have adopted AI but did not announce the initiative in the press; thus, such firms are not included in this study. Moreover, although we have carefully filtered out the confounding events within a five-day window, we are unable to consider all of the confounding factors in the research. For instance, we have not considered the confounding events among companies (Dardan et al. 2005). Finally, the research only examines the short-term impacts of AI investments. Examining the long-term effects of AI on operational performance will be worthwhile. The use of other methodologies, including surveys and case studies, to explore the effects of AI initiatives may also be considered.

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References

- Aouadni, I., & Rebai, A. (2017). Decision support system based on genetic algorithm and multi-criteria satisfaction analysis (MUSA) method for measuring job satisfaction. *Annals of Operations Research*, 256(1), 3–20.
- Bannerjee, G., Sarkar, U., Das, S., & Ghosh, I. (2018). Artificial intelligence in agriculture: A literature survey. *International Journal of Scientific Research in Computer Science Applications and Management Studies*, 7(3), 1–6.
- Basri, W. (2020). Examining the impact of artificial intelligence (AI)-assisted social media marketing on the performance of small and medium enterprises: toward effective business management in the Saudi Arabian context. *International Journal of Computational Intelligence Systems*, 13(1), 142–152.
- Benotmane, R., Kovács, G., & Dudás, L. (2019). Economic, social impacts and operation of smart factories in industry 4.0 focusing on simulation and artificial intelligence of collaborating robots. *Social Sciences*, 8(5), 143.
- Bharadwaj, A. S. (2000). A resource-based perspective on information technology capability and firm performance: an empirical investigation. *MIS Quarterly*, 24(1), 169–196.
- Bose, I., & Leung, A. C. M. (2013). The impact of adoption of identity theft countermeasures on firm value. *Decision Support Systems*, 55(3), 753–763.
- Bose, I., & Leung, A. C. M. (2014). Do phishing alerts impact global corporations? A firm value analysis. *Decision Support Systems*, 64, 67–78.
- Bose, I., Lui, A. K. H., & Ngai, E. W. T. (2011). The impact of RFID adoption on the market value of firms: An empirical analysis. *Journal of Organizational Computing and Electronic Commerce*, 21(4), 268–294.
- Bose, I., & Pal, R. (2012). Do green supply chain management initiatives impact stock prices of firms? *Decision Support Systems*, 52(3), 624–634.
- Bughin, J., Seong, J., Manyika, J., Chui, M., & Joshi, R. (2018). Notes from the AI frontier: Modeling the impact of AI on the world economy. *McKinsey Global Institute*.
- Castellano, R., & D'Ecclesia, R. L. (2013). CDS volatility: The key signal of credit quality. *Annals of Operations Research*, 205(1), 89–107.
- Castellanos, S. (2018). Bank of America tech chief defines responsible AI projects. *The Wall Street Journal* Retrieved July 3, 2020 from <https://blogs.wsj.com/cio/2018/12/05/bank-of-america-tech-chief-defines-responsible-ai-projects/>.
- Chatterjee, D., Pacini, C., & Sambamurthy, V. (2002). The shareholder-wealth and trading-volume effects of information-technology infrastructure investments. *Journal of Management Information Systems*, 19(2), 7–42.
- Chen, T.-L., Cheng, C.-Y., & Chou, Y.-H. (2020). Multi-objective genetic algorithm for energy-efficient hybrid flow shop scheduling with lot streaming. *Annals of Operations Research*, 290, 813–836.
- Chetthamrongchai, P., & Jermittiparsert, K. (2020). The impact of artificial intelligence outcomes on the performance of pharmacy business in Thailand. *Systematic Reviews in Pharmacy*, 11(1), 139–148.
- Chou, J.-S., Tai, Y., & Chang, L.-J. (2010). Predicting the development cost of TFT-LCD manufacturing equipment with artificial intelligence models. *International Journal of Production Economics*, 128(1), 339–350.
- Chui, M., Henke, N., & Miremadi, M. (2018). *Most of AI's business uses will be in two areas*. Retrieved July 3, 2020 from <https://hbr.org/2018/07/most-of-ais-business-uses-will-be-in-two-areas>.
- Dardan, M., Stylianou, A., & Dardan, S. (2005). The valuation of eCommerce announcements during fluctuating financial markets. *Journal of Electronic Commerce Research*, 6(4), 312–326.
- Dehning, B., Richardson, V. J., & Zmud, R. W. (2003). The value relevance of announcements of transformational information technology investments. *MIS Quarterly*, 27(4), 637–656.
- Do, N. A. D., Nielsen, I. E., Chen, G., & Nielsen, P. (2016). A simulation-based genetic algorithm approach for reducing emissions from import container pick-up operation at container terminal. *Annals of Operations Research*, 242(2), 285–301.
- Dos Santos, B. L., Peffers, K., & Mauer, D. C. (1993). The impact of information technology investment announcements on the market value of the firm. *Information Systems Research*, 4(1), 1–23.
- Dosdoğru, A. T., Boru İpek, A., & Göçken, M. (2020). A novel hybrid artificial intelligence-based decision support framework to predict lead time. *International Journal of Logistics Research and Applications*. <https://doi.org/10.1080/13675567.2020.1749249>.
- Faggella, D. (2019). *How investors feel about artificial intelligence—From 29 AI founders and executives*. Retrieved May 14, 2020 from <https://emerj.com/ai-market-research/how-investors-feel-about-artificial-intelligence-from-29-ai-founders-and-executives/>.

- Fan, W., Liu, J., Zhu, S., & Pardalos, P. M. (2020). Investigating the impacting factors for the healthcare professionals to adopt artificial intelligence-based medical diagnosis support system (AIMDSS). *Annals of Operations Research*, 294, 567–592.
- Fethi, M. D., & Pasiouras, F. (2010). Assessing bank efficiency and performance with operational research and artificial intelligence techniques: A survey. *European Journal of Operational Research*, 204(2), 189–198.
- Fragapane, G., Ivanov, D., Peron, M., Sgarbossa, F., & Strandhagen, J. O. (2020). Increasing flexibility and productivity in Industry 4.0 production networks with autonomous mobile robots and smart intralogistics. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-020-03526-7>.
- Hayes, D. C., Hunton, J. E., & Reck, J. L. (2001). Market reactions to ERP implementation announcements. *Journal of Information Systems*, 15(1), 3–18.
- Hunter, S. D. (2003). Information technology, organizational learning, and the market value of the firm. *Journal of Information Technology Theory and Application*, 5(1), 1–28.
- Im, K. S., Dow, K. E., & Grover, V. (2001). A reexamination of IT investment and the market value of the firm—An event study methodology. *Information Systems Research*, 12(1), 103–117.
- Jain, V. (2019). An impact of artificial intelligence on business. *International Journal of Research and Analytical Reviews*, 6(2), 302–308.
- Jiang, F., Jiang, Y., Zhi, H., Dong, Y., Li, H., Ma, S., et al. (2017). Artificial intelligence in healthcare: Past, present and future. *Stroke and Vascular Neurology*, 2(4), 230–243.
- Joyce, L. (2018). *Artificial intelligence and the banking industry's \$1 trillion opportunity*. Retrieved Jun 4, 2020 from <https://thefinancialbrand.com/72653/artificial-intelligence-trends-banking-industry/>.
- Kalayci, C. B., Polat, O., & Gupta, S. M. (2016). A hybrid genetic algorithm for sequence-dependent disassembly line balancing problem. *Annals of Operations Research*, 242(2), 321–354.
- Karimi, J., Somers, T. M., & Bhattacharjee, A. (2007). The role of information systems resources in ERP capability building and business process outcomes. *Journal of Management Information Systems*, 24(2), 221–260.
- Kobbacy, K. A., & Vadera, S. (2011). A survey of AI in operations management from 2005 to 2009. *Journal of Manufacturing Technology Management*, 22(6), 706–733.
- Kobbacy, K. A., Vadera, S., & Rasmy, M. H. (2007). AI and OR in management of operations: history and trends. *Journal of the Operational Research Society*, 58(1), 10–28.
- Kohli, R., & Devaraj, S. (2003). Measuring information technology payoff: A meta-analysis of structural variables in firm-level empirical research. *Information Systems Research*, 14(2), 127–145.
- Lam, H. K., Zhan, Y., Zhang, M., Wang, Y., & Lyons, A. (2019). The effect of supply chain finance initiatives on the market value of service providers. *International Journal of Production Economics*, 216, 227–238.
- Lawrynowicz, A. (2011). Advanced scheduling with genetic algorithms in supply networks. *Journal of Manufacturing Technology Management*, 22(6), 748–769.
- Lim, J.-H., Stratopoulos, T. C., & Wirjanto, T. S. (2013). Sustainability of a firm's reputation for information technology capability: The role of senior IT executives. *Journal of Management Information Systems*, 30(1), 57–96.
- Lu, Y., & Ramamurthy, K. (2011). Understanding the link between information technology capability and organizational agility: An empirical examination. *MIS Quarterly*, 35(4), 931–954.
- Makridakis, S. (2017). The forthcoming artificial intelligence (AI) revolution: Its impact on society and firms. *Futures*, 90, 46–60.
- Marr, B. (2019). *The 10 best examples of how companies use artificial intelligence in practice*. Retrieved Jun 3, 2020 from <https://www.forbes.com/sites/bernardmarr/2019/12/09/the-10-best-examples-of-how-companies-use-artificial-intelligence-in-practice/#5591731a7978>.
- McNaught, K., & Chan, A. (2011). Bayesian networks in manufacturing. *Journal of Manufacturing Technology Management*, 22(6), 734–747.
- Minevich, M. (2020). *4 Ways that you can prove ROI from AI*. Retrieved May 15, 2020 from <https://www.forbes.com/sites/markminevich/2020/03/03/4-ways-that-you-can-prove-roi-from-ai/#208b5fa784a7>.
- Mueller, J. P., & Massaron, L. (2018). *Artificial intelligence for dummies*. Hoboken: Wiley.
- Muhanna, W. A., & Stoel, M. D. (2010). How do investors value IT? An empirical investigation of the value relevance of IT capability and IT spending across industries. *Journal of Information Systems*, 24(1), 43–66.
- Munguia, J., Lloveras, J., Llorens, S., & Laoui, T. (2010). Development of an AI-based rapid manufacturing advice system. *International Journal of Production Research*, 48(8), 2261–2278.
- Murry, A. (2017). *Fortune 500 CEOs see A.I. As a big challenge*. Retrieved May 18, 2020 from <https://fortune.com/2017/06/08/fortune-500-ceos-survey-ai/>.
- Oh, W., Kim, J. W., & Richardson, V. J. (2006). The moderating effect of context on the market reaction to IT investments. *Journal of Information Systems*, 20(1), 19–44.

- Ranganathan, C., & Brown, C. V. (2006). ERP investments and the market value of firms: Toward an understanding of influential ERP project variables. *Information Systems Research*, 17(2), 145–161.
- Ross, J. W., Beath, C. M., & Goodhue, D. L. (1996). Develop long-term competitiveness through IT assets. *Sloan Management Review*, 38(1), 31–42.
- Sohal, A. S., Moss, S., & Ng, L. (2001). Comparing IT success in manufacturing and service industries. *International Journal of Operations & Production Management*, 21(1/2), 30–45.
- Statista. (2016). *Enterprise artificial intelligence market revenue worldwide 2016–2025*. Retrieved May 18, 2020 from <https://www.statista.com/statistics/607612/worldwide-artificial-intelligence-for-enterprise-applications/>.
- Staw, B. M., Sandelands, L. E., & Dutton, J. E. (1981). Threat rigidity effects in organizational behavior: A multilevel analysis. *Administrative Science Quarterly*, 26(4), 501–524.
- Stoel, M. D., & Muhanna, W. A. (2009). IT capabilities and firm performance: A contingency analysis of the role of industry and IT capability type. *Information & Management*, 46(3), 181–189.
- Subramani, M., & Walden, E. (2001). The Impact of e-Commerce announcements on the market value of firms. *Information Systems Research*, 12(2), 135–154.
- Tanriverdi, H., & Ruefli, T. W. (2004). The role of information technology in risk/return relations of firms. *Journal of the Association for Information Systems*, 5(11–12), 421–447.
- Teo, T. S., Nishant, R., & Koh, P. B. (2016). Do shareholders favor business analytics announcements? *The Journal of Strategic Information Systems*, 25(4), 259–276.
- White, L. J. (2010). Markets: The credit rating agencies. *Journal of Economic Perspectives*, 24(2), 211–226.
- Yang, L., Chen, G., Rytter, N. G. M., Zhao, J., & Yang, D. (2019). A genetic algorithm-based grey-box model for ship fuel consumption prediction towards sustainable shipping. *Annals of Operations Research*. <https://doi.org/10.1007/s10479-019-03183-5>.

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